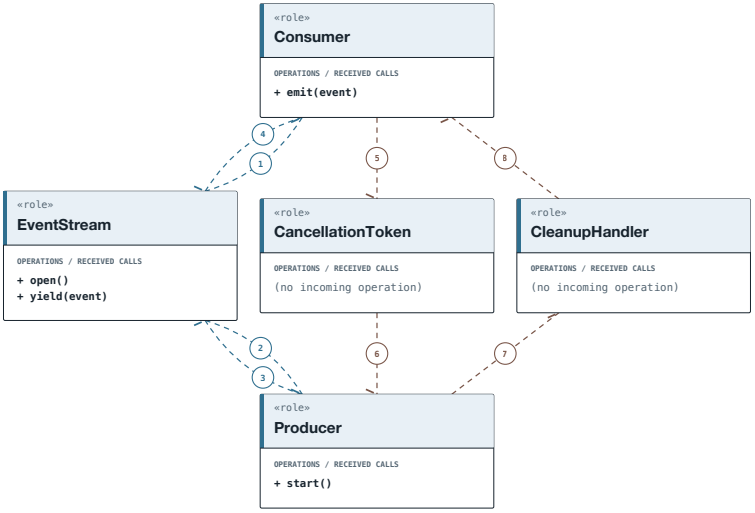
P05 Asynchronous-generator loop

Primitive / UML CLASS VIEW / ANALYTICAL MODEL

Expose incremental execution events.

UML CLASS VIEW

dashed arrow: call dependency · brown: return/failure · analytical, not an observed trace



POINTCUT / PATTERN INTERFACE

CONTEXT	A consumer observes an agent run incrementally.
TRIGGER	The runtime emits a semantic event or streamed token.
STRUCTURE	Async yield/resume, cancellation, and cleanup ownership.
EFFECT	Consumers receive ordered events and early exit releases runtime resources.

CONTRACT / REQUIRED BEHAVIOR

REQUIRES	The producer exposes an async iterator or equivalent streaming boundary.
ENSURES	Each emitted event has a stable type and sequence context.
INVARIANT	Cancellation and producer failure do not leave unreleased work.
VIOLATION	Abandoned consumers trigger cancellation or documented cleanup.

RELATIONSHIPS / ANALYTICAL CALLS

number · caller → callee · message

1 Consumer → EventStream open()
2 EventStream → Producer start()
3 Producer → EventStream yield(event)
4 EventStream → Consumer emit(event)
5 Consumer ↩ CancellationToken cancel()
6 CancellationToken ↩ Producer cancel()
7 Producer ↩ CleanupHandler release(next)
8 CleanupHandler ↩ Consumer closed()

VARIATION / TEST / NEGATIVE CONTROL

VARIATION	Stream event schema, backpressure, buffering, and consumer adapters.
TEST_OBLIGATION	Test event order, consumer cancellation, producer exception, and cleanup.
NEGATIVE_CONTROL	A synchronous list returned at the end is not the streaming pointcut.